SUPPORTING ADULT GUIDANCE

Rounding: which is nearest?

Place Value · Lesson 7. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

In the main activity, the Nearest Marker, a number sits on a line between two markers with the halfway point shown. Learners slide the number, type one, or move the window, and choose what to round to (1,000, 100, 10, whole number or 0.1). The nearer marker lights up and the distances to each marker are shown; learners tap the marker they think the number rounds to. Seven rounding questions include 345 (halfway) and 4.97 to the nearest 0.1 (5.0). Six questions follow, and a stretch explains why the "5 or more" shortcut works. Learners then round quickly, judge statements (including the "rounding twice" error), and work backwards from a rounded answer.

Driving Question: *Which multiple of 10, 100 or 0.1 is this number closest to?*

The mathematics, in plain terms

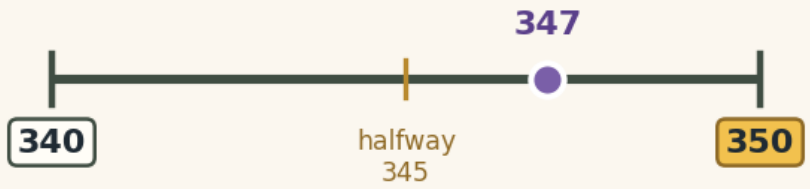
Rounding to the nearest multiple of 10■ chooses the nearer of the two neighbouring multiples. Exact halfway cases are settled by convention; this lesson uses "round half up", the convention expected in school mathematics. The "look at the next digit" rule is equivalent: a next digit of 5 to 9 means halfway or beyond. Answers to the nearest 0.1 are written with one decimal place (5.0, not 5), though 5 is accepted as the same value. Significant figures and error intervals build on this later.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — 347 to the nearest 10

Round 347 to the nearest 10.

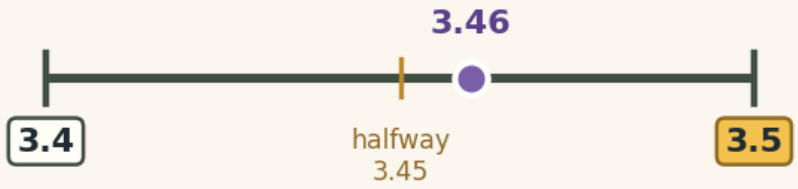


1. 347 lies between the markers	340 and 350
2. Halfway between them is	345
3. 347 is past halfway: 3 from 350, 7 from 340.	350

347 rounded to the nearest 10 is 350.

Example 2 — 3.46 to the nearest 0.1

Round 3.46 to the nearest 0.1.



1. 3.46 lies between the markers	3.4 and 3.5
2. Halfway between them is	3.45
3. 3.46 is past halfway: 0.04 from 3.5, 0.06 from 3.4.	3.5

3.46 rounded to the nearest 0.1 is 3.5.

See the nearer marker, then explain the shortcut

Learners decide by distance on the line before meeting any rule, so the halfway point is something they can see. The stretch then shows that "look at the next digit" is a quick way of asking which side of halfway the number is. The sense check deliberately includes rounding in steps, a common error.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The Nearest Marker and six questions. Stretch: why does "look at the next digit" work?	~16 min
Movement	Round it: eight quick roundings.	~7 min
Reflection	Sense check: true or not?	~8 min
Creativity	What could it be? Work backwards from a rounded answer.	~10 min
Rest	Close enough is a choice.	~2 min
Nutrition	Intellectual: estimation.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Rounding in steps: 349 → 350 → 400 to the nearest 100.
Ask: “Which two hundreds is 349 between? Which is nearer?”

Giving an answer that is not a marker (583 to the nearest 10 as 590 or 58).
Ask: “Which two tens is 583 between?”

Rounding 4.97 to the nearest 0.1 as 4.10.
Ask: “Which two tenths is 4.97 between? What comes after 4.9?”

Rounding a halfway number down.
Ask: “It is the same distance from both. What have we agreed to do then?”

Looking at the wrong digit (the last digit rather than the next column).
Ask: “Which column are you rounding to? Which column comes straight after it?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: comparing numbers (Lesson 6). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner rounds 2,549 to the nearest 100 and says 2,600, because 9 rounds up the 4. The aim is a question that brings back the line.

- “Which two hundreds is 2,549 between?”
- “Where is halfway between them?”

- “Is 2,549 before or after 2,550?”
- “Which digit tells you that straight away?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why “round half up”? Isn't 345 equally close to both?
Yes, so a convention is needed. Round half up is the convention used in school mathematics and examinations; other fields sometimes use other conventions.

Should learners write 5.0 or 5 for 4.97 to the nearest 0.1?
5.0 shows the degree of accuracy, so encourage it. The lesson accepts 5 as the same value.

Is the stretch required?
No. The secure core is rounding to the nearest 10, 100, 1,000, whole number and 0.1, explained with the halfway point.

What comes next?
Lesson 8: one number, many representations.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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